

GLA UNIVERSITY, MATHURA

B.Tech. II Year, III/IV Semester

Subject Name-Probability and Statistics

Assignment-02

- 1) The probability of a man hitting the target is $\frac{1}{4}$.
 - (i) If he fires 7 times, what is the probability that he hits the target:
 - (a) at least once,
 - (b) at least twice
 - (ii) How many times must be fired so that the probability of his hitting the target at least once is greater than $\frac{2}{3}$?
- 2) A light bulb manufacturing factory finds 3 in every 60 light bulbs defective. What is the probability that the first defective light bulb will be found when the 6th one is tested?
- 3) A manufacturer who produces medicine bottles finds that 0.1% of the bottles are defective. The bottles are packed in boxes, each containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain (i) no defective bottles, (ii) at least two defective bottles. (given $e^{-0.5} = 0.6065$).
- 4) If a random variable X follows a Poisson distribution such that $P(X = 2) = 9P(X = 4) = 90P(X = 6)$, Find the mean and variance of X .
- 5) In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.
- 6) The marks obtained in a certain examination follow normal distribution with mean 45 and standard deviation 10. If 1000 students appeared at the examination, calculate the number of students scoring.
 - (i) Less than 40 marks,
 - (ii) More than 60 marks.

[Given $P(Z > 0.5) = 0.3085, P(Z < -1.5) = 0.9332$]
- 7) The local authorities in a certain city install 10000 electric lamps in the streets of Mathura city. If these lamps have an average life of 1000 burning hours with a standard deviation of 200 hours. Assuming the lifetime of a lamp follows Normal Distribution, what number of lamps might be expected to fail (i) in the first 800 burning hours, and (iii) between 800 and 1200 burning hours? [Given $P(0 < Z < 1) = 0.3413$]

8) Calculate the coefficient of correlation between the following data:

X: 10,000 20,000 30,000 40,000 50,000 60,000 70,000
 Y: 0.3 0.5 0.6 0.8 1.0 1.1 1.3

9) Calculate the Karl Pearson's coefficient between x and y for the following information.

$$n = 12, \sum x = 120, \sum (x - 8)^2 = 150, \sum y = 130, \\ \sum (y - 10)^2 = 200 \text{ And } \sum (x - 8)(y - 10) = 50$$

10) The following table gives the ages and blood pressure of 10 women:

Age(X)	:	56	42	36	47	49	42	60	72	63	55
Blood Pressure(Y)	:	147	125	118	128	145	140	155	160	149	150

- (i) Find the correlation coefficient between X and Y.
- (ii) Determine the regression equation of Y on X.
- (iii) Estimate the blood pressure of a woman whose age is 45 years.

11) Regression equations of two variables X and Y are as follows:

$$8X - 10Y = 64 \text{ and } 40X - 18Y = 320$$

Find (i) The means, (ii) The regression coefficients, and (iii) The coefficients of correlation between X and Y.

12) A certain drug is claimed to be effective in curing colds. In an experiment on 500 persons with cold, half of them were given the drug and half of them were given sugar pills. The patients' reactions to the treatment are recorded in the following table:

Treatment	Helped	Reaction	No effect	Total
Drug	150	30	70	250
Sugar Pills	130	40	80	250
Total	280	70	150	500

On the basis of data, can it be concluded that there is a significance difference in the effect of the drug and sugar pills? [Given $\chi^2_{2,0.05} = 5.99$]

13) In previous years, the marks obtained in a French test by students attending Topnotch College have been modelled satisfactorily by a normal distribution with mean of 65 and a standard deviation of 9. Teachers in the French department at Topnotch college suspect that this year their students are, on average, underachieving. In order to investigate this suspicion, the teachers selected a random sample of 25 students to take the French test and found that their mean score was 61.5. [$t_{24,0.05} = -1.712$]

- a) Investigate, at the 5% level of significance, the teachers' suspicion.
- b) Explain, in the context of this question, the meaning a Type I error.

14) A company has reorganized its sales department. The following data show its weekly sales both before and after reorganization. The period for comparison is taken from January to march in two successive years.

Week No.	1	2	3	4	5	6	7	8	9	10
Sales prior to reorganization	12	15	13	11	17	15	10	11	18	19
Sales after to reorganization	16	17	14	13	15	14	12	11	17	22

Can it be concluded that the reorganization of the sales department of the company has resulted in a significant increase in its sales. [$t_{18,0.05} = 1.734$]

15) Resting heart rate is known to be 71 beats per minute on average, with a standard deviation of 4 beats per minute. A set of researchers believe that heart rate will increase in men when they are waiting to go in to a job interview. To test this hypothesis, a group of 9 men attending job interviews are fitted with a wireless heart rate monitor to wear on their chest in the hour preceding their interviews. Their average heart rates over this hour are shown in the table below.

No. of participant	Participant Heart rate (bmp)
1	80
2	74
3	73
4	72
5	78
6	75
7	70
8	74
9	69

- a) Is significant evidence to suggest heart rate increases in men while they are waiting to attend a job interview? Conduct the test at the 5% level and interpret your result. [$t_{8,0.05} = 1.859$]